

CLEAR

Clean, Leakage-free Evaluation And Reporting
an honest evaluation harness for highway trajectory prediction

1 INPUTS

- **A dataset**

highD · exiD · NGSIM

you provide it — CLEAR ships none

- **Your model**

subclass Predictor

predict(ws) → (N, T, 2)

2 CLEAR PIPELINE

- **Canonical frame**

per-window, handles ramps

- **Leakage-safe split**

+ PASS / WARN verdict

- **CV · CA · IDM**

honest baselines

- **Metrics**

FDE/RMSE · CIs · CA win-rate

strata · best-of-k · collisions

3 EVAL CARD

- **Protocol**

+ leakage verdict

- **Accuracy ladder**

every model vs CA

- **Stratified error**

cruise / mild / hard

- **Admissibility**

predicted collisions

- **5-point checklist**

JSON · Markdown · figure

WHY IT MATTERS

- **Leakage inflates.**

Sliding windows share vehicles and recordings across folds; CLEAR forbids it by default.

- **Averaging hides the tail.**

~77% of windows are trivial cruising; one RMSE buries the safety-critical cases.

- **CA is a hard baseline.**

A zero-parameter constant-acceleration model is unbeaten on clean highway data.

- **Accuracy ≠ safety.**

The lowest-error model can forecast the most rear-end collisions — CLEAR measures both.